

Sketch proofs of the ContEst appendix regularity propositions

ContEst supplementary note

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Abstract

This note gives sketch proofs of the two regularity results proved in the appendices of the ContEst paper: the LQG regularity proposition (Appendix A), under which the covariance-SDP design gradients of the LQG inner problem are exact, and the fixed- γ H_∞ regularity proposition (Appendix B), its robust twin for the game-Riccati inner problem. For each we state the claim, then argue at the level of the decisive identities and the classical theorems they invoke — Slater/conic duality, the discrete algebraic Riccati equation (DARE) and its game analog (GARE), the implicit function theorem via the Stein operator, the Danskin/Bonnans–Shapiro value-derivative formula, and the Alizadeh–Haeberly–Overton characterization of nondegeneracy — explaining briefly why each applies. The fully expanded LQG argument is in the companion `Thm1_proof.tex`; this note is the compact version and additionally covers H_∞ .

1 Common setup and notation

Fix a design point θ in an open set $\Theta_0 \subseteq \mathbb{R}^{r_\theta}$ and the discrete-time model

$$x_{k+1} = A_\theta x_k + B_\theta u_k + w_k, \quad y_k = C_\theta x_k + v_k, \quad (1)$$

$x_k \in \mathbb{R}^n$, $u_k \in \mathbb{R}^m$, $y_k \in \mathbb{R}^p$, with C^1 data $\theta \mapsto (A_\theta, B_\theta, C_\theta, W, V, Q, R)$, noise covariances W, V and stage weights Q, R . We write $\mathbb{S}^q$ for symmetric $q \times q$ matrices, $\langle X, Y \rangle = \text{tr}(X^\top Y)$, and call M *Schur* if $\rho(M) < 1$. Both inner problems are instances of the paper’s generic convex program

$$V(\theta) = \min_z \phi(z, \theta) \quad \text{s.t.} \quad g(z, \theta) = 0, \quad G(z, \theta) \in \mathcal{K}, \quad (2)$$

with $\mathcal{K}$ the PSD cone, Lagrangian $L = \phi + \langle \mu, g \rangle + \langle S, G \rangle$, optimal triple (z^*, μ^*, S^*) . The paper’s Theorem 1 (envelope theorem) states that under its Assumption 1,

$$\nabla_\theta V = \partial_\theta \phi + \langle \mu^*, \partial_\theta g \rangle + \langle S^*, \partial_\theta G \rangle, \quad (3)$$

the partials taken with (z^*, μ^*, S^*) frozen. Assumption 1 asks, on Θ_0 : (A1) convexity, C^1 data, Slater; (A2) unique primal minimizer; (A3) nondegeneracy/strict complementarity, hence a unique dual; (A4) θ enters only through finitely many C^1 data objects. The two propositions below reduce (A1)–(A4) to *system-theoretic* hypotheses that are checkable from the model.

Throughout, the standing system-theoretic hypotheses are

- (S1) the data maps are C^1 in θ ;
- (S2) (A_θ, B_θ) stabilizable and (A_θ, C_θ) detectable;
- (S3) $W, V, R \succ 0$, $Q \succeq 0$ with $(A_\theta, Q^{1/2})$ detectable (automatic when $Q \succ 0$).

2 Appendix A: LQG regularity — sketch

The LQG inner value splits, by the fixed- θ separation principle, into a control SDP and an estimation SDP. In covariance form the control SDP is

$$J_{\text{cont}}(\theta) = \min_{\Sigma, L, Z_0} \text{tr}(Q\Sigma) + \text{tr}(RZ_0) \quad \text{s.t.} \quad \Sigma \succ 0, \begin{bmatrix} Z_0 & L \\ L^\top & \Sigma \end{bmatrix} \succeq 0, \mathcal{G}_c \succeq 0, \quad (4)$$

with $\mathcal{G}_c = \begin{bmatrix} \Sigma - W & A_\theta \Sigma + B_\theta L \\ (A_\theta \Sigma + B_\theta L)^\top & \Sigma \end{bmatrix}$ and optimal gain $K^* = L^*(\Sigma^*)^{-1}$; the estimation SDP is its transpose dual (Section 2.1).

Proposition 1 (LQG regularity). *Under (S1)–(S3), for (4) and its estimation dual and every $\theta \in \Theta_0$: (a) Slater holds; (b) the primal minimizer is unique and equals the stabilizing DARE solution, $\Sigma^* \succ 0$; (c) the value is C^1 and, for every optimal dual, (3) returns $\nabla_\theta V$ (so the design gradients read from the solver’s duals are exact); (d) if in addition the solution is strictly complementary and primal-nondegenerate, the dual is unique and Assumption 1 holds verbatim. Moreover the (1,1) dual block of $\mathcal{G}_c$ equals the control Riccati solution, $S_{11}^* = P$.*

(a) Slater from stabilizability/detectability. Pick K with $A_{\text{cl}} := A_\theta + B_\theta K$ Schur (exists by (S2)). For any $\varepsilon > 0$ the discrete Lyapunov equation $\Sigma = A_{\text{cl}} \Sigma A_{\text{cl}}^\top + W + \varepsilon I$ has a unique solution $\Sigma = \sum_{k \geq 0} A_{\text{cl}}^k (W + \varepsilon I) A_{\text{cl}}^{\top k} \succ 0$ because A_{cl} is Schur and $W + \varepsilon I \succ 0$ (Lyapunov stability theorem for discrete systems). Setting $L = K\Sigma$, $Z_0 = K\Sigma K^\top + \varepsilon I$, the Schur complements of the three LMIs are $\varepsilon I \succ 0$, so the point is *strictly* feasible. Slater for a bounded-below convex conic program gives zero duality gap and dual attainment by the *conic strong-duality theorem* [3, Thm. 2.4.1], [4, §5.9]. This is (A1); (A4) is immediate since θ enters only through $(A_\theta, B_\theta, W, Q, R)$, all C^1 .

(b) Unique primal = DARE solution. Two classical facts do the work. First, the covariance SDP is an exact convex lift of the stationary LQR problem: for feasible (Σ, L, Z_0) with $K = L\Sigma^{-1}$, the Schur complement of $\mathcal{G}_c$ reads $\Sigma \succeq A_{\text{cl}} \Sigma A_{\text{cl}}^\top + W$, and $W \succ 0$ forces A_{cl} Schur (a *Lyapunov-inequality / eigenvector* argument [5, Ch. 5]); iterating the resulting Stein inequality gives $\Sigma \succeq \Sigma_K$, so the objective dominates the stationary LQR cost $J(K) = \text{tr}(Q\Sigma_K) + \text{tr}(RK\Sigma_K K^\top)$. Second, the *completion-of-squares identity*

$$J(K) - \text{tr}(PW) = \text{tr}((K - K^*)^\top (R + B_\theta^\top P B_\theta) (K - K^*) \Sigma_K), \quad (5)$$

with P the stabilizing solution of the DARE $P = A_\theta^\top P A_\theta - A_\theta^\top P B_\theta (R + B_\theta^\top P B_\theta)^{-1} B_\theta^\top P A_\theta + Q$, is ≥ 0 and vanishes iff $K = K^*$ (here $R + B_\theta^\top P B_\theta \succ 0$, $\Sigma_K \succ 0$). Existence/uniqueness of the stabilizing $P \succeq 0$ is the *DARE theorem* under (S2)–(S3) [7, Ch. 14–15], [6, Ch. 6]. Hence the minimizer $\Sigma^* = \Sigma_{K^*} \succ 0$, $L^* = K^* \Sigma^*$ is unique and $J_{\text{cont}} = \text{tr}(PW)$, which is (A2). (For $Q \succeq 0$ the final pinning of Σ^* uses $(A_\theta, Q^{1/2})$ detectability; for $Q \succ 0$ it is immediate.)

(c) C^1 value and dual-invariant gradient (the operational core). Because $J_{\text{cont}} = \text{tr}(PW)$ and $P(\theta)$ is a C^1 function of the data, the value is C^1 : apply the *implicit function theorem* to the DARE residual $\mathcal{R}(P, \theta) = 0$, whose P -derivative at a stabilizing root is the Stein operator $\delta P \mapsto \delta P - A_{\text{cl}}^* \delta P A_{\text{cl}}^*$, invertible exactly because A_{cl}^* is Schur (eigenvalues $1 - \lambda_i \bar{\lambda}_j \neq 0$). Now the key point that makes the gradient safe to read off *any* returned dual: by the *Danskin/Bonnans–Shapiro directional-derivative theorem* for optimal values of convex programs with a unique primal solution [3, Thm. 4.24],

$$V'(\theta; d) = \min_{(\mu, S) \in \mathcal{D}^*(\theta)} \langle \nabla_\theta L(z^*, \mu, S, \theta), d \rangle,$$

where $\mathcal{D}^*$ is the (compact convex) dual optimal set. Since V is C^1 , the left side is *linear* in d , so the min equals $\langle \nabla_\theta V, d \rangle$ for all d ; a fixed dual gives $\langle \nabla_\theta L, d \rangle \geq \langle \nabla_\theta V, d \rangle$ for all d (including $-d$), forcing equality $\nabla_\theta L = \nabla_\theta V$. Thus *every* optimal dual yields the same, correct gradient, and Corollary 1 of the paper holds under (S1)–(S3) *without* strict complementarity. Specialized to the Lyapunov LMI, $\nabla_\theta L$ is the paper’s $\partial J_{\text{cont}}/\partial A_\theta = -2S_{12}\Sigma^*$, etc.

The dual identity. Complementary slackness aligns range S^* with $\ker \mathcal{G}_c^* = \text{range}[I; -A_{\text{cl}}^{*\top}]$ (the tight Lyapunov block has rank n), so $S^* = [I; -A_{\text{cl}}^{*\top}]M[I; -A_{\text{cl}}^{*\top}]^\top$ and $S_{12}^* = -S_{11}^*A_{\text{cl}}^*$. Lagrangian stationarity in Σ then gives the closed-loop *Stein equation* $S_{11} = Q + K^{*\top}RK^* + A_{\text{cl}}^{*\top}S_{11}A_{\text{cl}}^*$, whose unique solution (A_{cl}^* Schur) is P ; hence $S_{11}^* = P$ and, via $S_{12}^* = -PA_{\text{cl}}^*$, $\partial J_{\text{cont}}/\partial A_\theta = -2S_{12}^*\Sigma^* = 2PA_{\text{cl}}^*\Sigma^*$, matching the Riccati gradient. (This is the identity corrected in the paper’s Appendix A: the off-diagonal dual block carries A_{cl}^* , not $K^{*\top}$.)

(d) Strict complementarity $\Rightarrow$ unique dual. Primal nondegeneracy at z^* implies the dual optimizer is unique and strict complementarity implies the primal–dual solution map is differentiable, by the *Alizadeh–Haeberly–Overton / Shapiro* characterization for SDPs [1, Thms. 10,15],[2, Thm. 4.1]. Here $\text{rank } \mathcal{G}_c^* = n$ always, so strict complementarity reduces to $\text{rank } S^* = n$ with $S^*\mathcal{G}_c^* = 0$, guaranteed by rank counting from $P \succeq Q \succ 0$, $R \succ 0$. This supplies (A3), completing Assumption 1.

2.1 The estimation SDP by transpose duality

The estimation SDP is (4) under the relabeling $A_\theta \mapsto A_\theta^\top$, $B_\theta \mapsto C_\theta^\top$, $W \mapsto Q$, $Q \mapsto W$, $R \mapsto V$, $(\Sigma, L, Z_0) \mapsto (P^\varepsilon, -F, Z_1)$. The hypotheses map correctly: (A_θ, B_θ) stabilizable $\leftrightarrow (A_\theta, C_\theta)$ detectable, and $(A_\theta, Q^{1/2})$ detectable $\leftrightarrow (A_\theta, W^{1/2})$ stabilizable (automatic since $W \succ 0$ makes it controllable). Every step above is invariant under this relabeling, so all conclusions transfer, giving the unique $P^{\varepsilon*} = \Sigma^\varepsilon$ (Kalman error covariance) and $S_{11}^* = \Sigma^\varepsilon$.

3 Appendix B: fixed- γ H_∞ regularity — sketch

For the robust inner problem the paper uses, at a *fixed* attenuation γ , the game Riccati route. With disturbance channel $E_\theta = W_\theta^{1/2}$, augmented input $\tilde{B}_\theta = [B_\theta \ E_\theta]$ and *indefinite* game weight $\tilde{R} = \text{diag}(R, -\gamma^2 I)$, the guaranteed cost is $V_{\text{wc}}(\theta) = \text{tr}(XW_\theta)$ with $X \succeq 0$ the stabilizing solution of the game GARE

$$X = Q + A_\theta^\top X A_\theta - A_\theta^\top X \tilde{B}_\theta (\tilde{R} + \tilde{B}_\theta^\top X \tilde{B}_\theta)^{-1} \tilde{B}_\theta^\top X A_\theta, \quad (6)$$

saddle gain $\tilde{K} = -(\tilde{R} + \tilde{B}_\theta^\top X \tilde{B}_\theta)^{-1} \tilde{B}_\theta^\top X A_\theta$, closed loop $A_{\text{cl}} = A_\theta + \tilde{B}_\theta \tilde{K}$.

Proposition 2 (Fixed- γ H_∞ regularity). *Fix γ above the optimal attenuation and let $X \succeq 0$ solve (6) with $R + B_\theta^\top X B_\theta \succ 0$ and A_{cl} Schur. Then X is a locally unique C^1 function of θ , $V_{\text{wc}}(\theta) = \text{tr}(XW_\theta) \in C^1$, with the closed-form gradient (9) below. The dual filter GARE is regular under the analogous conditions, giving $V_{\text{est}}(\theta) = \text{tr}(M\Sigma_e) \in C^1$.*

The proof is the Riccati (implicit-function) half of Section 2, carried through with an indefinite weight; there is no SDP dual and no strict-complementarity requirement.

(a) Stabilizing solution. For γ above the optimal attenuation γ^* , the *discrete dynamic-game (bounded-real) Riccati theorem* gives a unique $X \succeq 0$ solving (6) with $R + B_\theta^\top X B_\theta \succ 0$, full pivot $\tilde{R} + \tilde{B}_\theta^\top X \tilde{B}_\theta$ nonsingular of inertia (m, r_w) , and A_{cl} Schur [8],[5, Ch. 2]. This is the fixed- γ analog of the DARE existence used in Section 2; approaching γ^* makes the pivot lose rank (the minimal- γ SDP degeneracy the paper avoids), so we stay strictly above it.

(b) Lyapunov (completion-of-squares) form. Substituting $\tilde{K}$ into $Q + A_{\text{cl}}^\top X A_{\text{cl}} + \tilde{K}^\top \tilde{R} \tilde{K}$ and cancelling yields

$$X = Q + \tilde{K}^\top \tilde{R} \tilde{K} + A_{\text{cl}}^\top X A_{\text{cl}}, \quad (7)$$

identical algebra to the LQG identity below (5); completion of squares never uses the sign of $\tilde{R}$, which is why indefiniteness is harmless.

(c) Local uniqueness and C^1 (the heart). The GARE residual $\mathcal{R}(X, \theta)$ (right minus left of (6)) is C^1 wherever the pivot is nonsingular, hence near a stabilizing root. Because $\tilde{K}$ is the *saddle point* of the inner game (nonsingular pivot), the game envelope annihilates the gain's first-order variation and the partial derivative in X collapses to the Stein operator

$$\partial_X \mathcal{R}(X, \theta)[\delta X] = A_{\text{cl}}^\top \delta X A_{\text{cl}} - \delta X, \quad (8)$$

invertible precisely because A_{cl} is Schur. The *implicit function theorem* then gives a unique C^1 branch $\theta \mapsto X(\theta)$ through the stabilizing root, and with it C^1 maps $\tilde{K}(\theta), A_{\text{cl}}(\theta)$ and the game gramian $S(\theta)$ solving $S = A_{\text{cl}} S A_{\text{cl}}^\top + W_\theta$. This is Proposition 2; note it needs no dual and no strict complementarity — the regularity is read straight off the Riccati map, unlike the SDP route of Section 2.

(d) C^1 value and gradient. $V_{\text{wc}} = \text{tr}(X W_\theta)$ is C^1 by (c) and (S1). Differentiating (7), saddle stationarity of $\tilde{K}$ kills all $d\tilde{K}$ terms, so dX solves the Stein equation $dX - A_{\text{cl}}^\top dX A_{\text{cl}} = \Xi$ with source $\Xi = dQ + \tilde{K}^\top d\tilde{R} \tilde{K} + A_{\text{cl}}^\top X (dA_\theta + d\tilde{B}_\theta \tilde{K}) + (\cdot)^\top$. Contracting with the gramian and using $\text{tr}(W_\theta dX) = \text{tr}(S(dX - A_{\text{cl}}^\top dX A_{\text{cl}})) = \text{tr}(S\Xi)$ gives, term by term,

$$\frac{\partial V_{\text{wc}}}{\partial A_\theta} = 2X A_{\text{cl}} S, \quad \frac{\partial V_{\text{wc}}}{\partial \tilde{B}_\theta} = 2X A_{\text{cl}} S \tilde{K}^\top, \quad \frac{\partial V_{\text{wc}}}{\partial Q} = S, \quad \frac{\partial V_{\text{wc}}}{\partial \tilde{R}} = \tilde{K} S \tilde{K}^\top, \quad (9)$$

plus the explicit $\partial V_{\text{wc}}/\partial W_\theta = X$ from the objective. The channel $E_\theta = W_\theta^{1/2}$ enters through the E -block of $\partial V_{\text{wc}}/\partial \tilde{B}_\theta$, so a design coordinate moving both the objective weight W_θ and the channel E_θ sums the two contributions (the data-object chain rule of Assumption 1(A4)).

(e) Filter GARE by duality; (f) the H_2 limit. Under the transpose substitution $A_\theta \mapsto A_\theta^\top$, $\tilde{B}_\theta \mapsto [C_\theta^\top \ L_e^\top]$, $\tilde{R} \mapsto \text{diag}(V, -\gamma^2 I)$, the worst-case error covariance Σ_e solves the control GARE on the dual data, so (a)–(d) transfer and $V_{\text{est}} = \text{tr}(M \Sigma_e) \in C^1$ with the dual gradient. Finally, as $\gamma \rightarrow \infty$ the $-\gamma^2 I$ pivot block deactivates, (6) reduces to the control DARE, $X \rightarrow P$, $S \rightarrow \Sigma^*$, and (9) collapses to the LQG gradients of Section 2: the H_2 /LQG result is the $\gamma \rightarrow \infty$ boundary of this one.

4 What each proof rests on

- **Slater (LQG only)** $\Leftarrow$ a stabilizing feedback plus an εI inflation, i.e. the controllability/observability data, with $W, V \succ 0$ supplying the strict margin; conic strong duality then gives (A1) [3, 4].

- **Primal uniqueness** $\Leftarrow$ the (game) Riccati equation: the covariance/certificate SDP is an exact convex lift whose optimizer is the (G)ARE solution [7, 6, 8], pinned by completion of squares.
- **Differentiability and gradient correctness** $\Leftarrow$ the *implicit function theorem* through the Stein operator (A_{cl} Schur), which alone makes the value C^1 ; and, in the SDP case, the *Danskin/Bonnans–Shapiro* formula, which makes the envelope gradient *dual-invariant* — so the returned-dual gradient is exact even without strict complementarity [3].
- **Dual uniqueness (A3)** $\Leftarrow$ strict complementarity / nondegeneracy in the *Alizadeh–Haeberly–Overton* sense [1, 2]; needed only for the “unique dual” upgrade, not for a correct gradient.
- **H_∞ vs. LQG.** The H_∞ proof is the Riccati half only: IFT + Stein operator + gradient, with an indefinite weight and no SDP dual. It therefore never needs Slater, dual invariance, or strict complementarity — the fixed- γ GARE is well-posed by construction, and the H_2 case is its $\gamma \rightarrow \infty$ limit.

References

- [1] F. Alizadeh, J.-P. A. Haeberly, and M. L. Overton, *Complementarity and nondegeneracy in semidefinite programming*, Math. Program. **77** (1997), 111–128.
- [2] A. Shapiro, *First and second order analysis of nonlinear semidefinite programs*, Math. Program. **77** (1997), 301–320.
- [3] J. F. Bonnans and A. Shapiro, *Perturbation Analysis of Optimization Problems*, Springer, New York, 2000.
- [4] S. Boyd and L. Vandenberghe, *Convex Optimization*, Cambridge University Press, 2004.
- [5] S. Boyd, L. El Ghaoui, E. Feron, and V. Balakrishnan, *Linear Matrix Inequalities in System and Control Theory*, SIAM, Philadelphia, 1994.
- [6] B. D. O. Anderson and J. B. Moore, *Optimal Control: Linear Quadratic Methods*, Prentice Hall, Englewood Cliffs, NJ, 1990.
- [7] P. Lancaster and L. Rodman, *Algebraic Riccati Equations*, Oxford University Press, 1995.
- [8] T. Başar and P. Bernhard, *H_∞ -Optimal Control and Related Minimax Design Problems: A Dynamic Game Approach*, 2nd ed., Birkhäuser, Boston, 1995.